



steamaco

Sell Power.
Profitably.



Our Vision: everyone has access
to reliable and affordable power



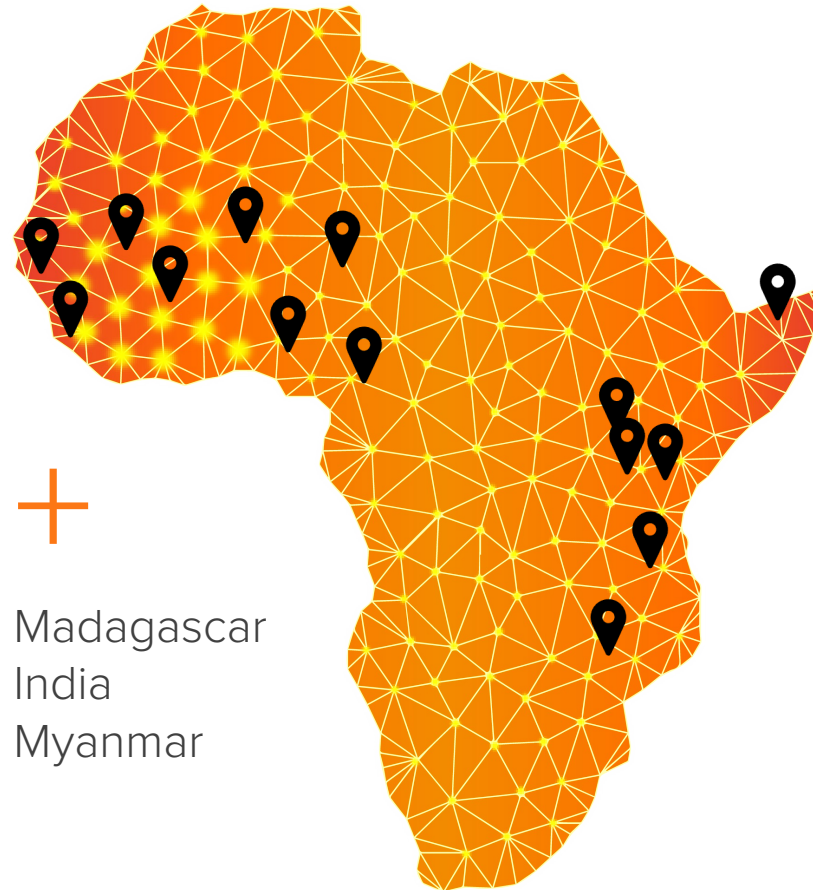
An aerial photograph of a rural village in a dry, hilly region. The village consists of numerous small, rectangular buildings with light-colored walls and dark roofs, scattered across a landscape of dry, brownish-yellow earth and sparse green trees. A prominent dirt road runs through the center of the village. In the foreground, a large, rectangular solar panel array is installed on a flat, open area, with a group of people gathered around it. The background shows rolling hills under a clear sky.

Our Vision: everyone has access
to reliable and affordable power

**Our Mission: help
utilities sell power
profitably**

Our presence:

Established tech partner to energy providers in Africa & Asia



30

Grid Operators

17

Countries

41k

Active Connections

96%

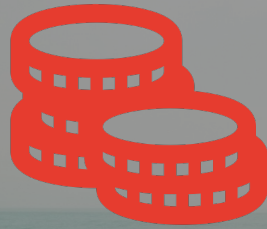
Average Uptime

Imagine if selling energy
was **easy**.



Imagine if selling energy was **easy**.

Are your collection processes risking your financial viability?



Selling enough



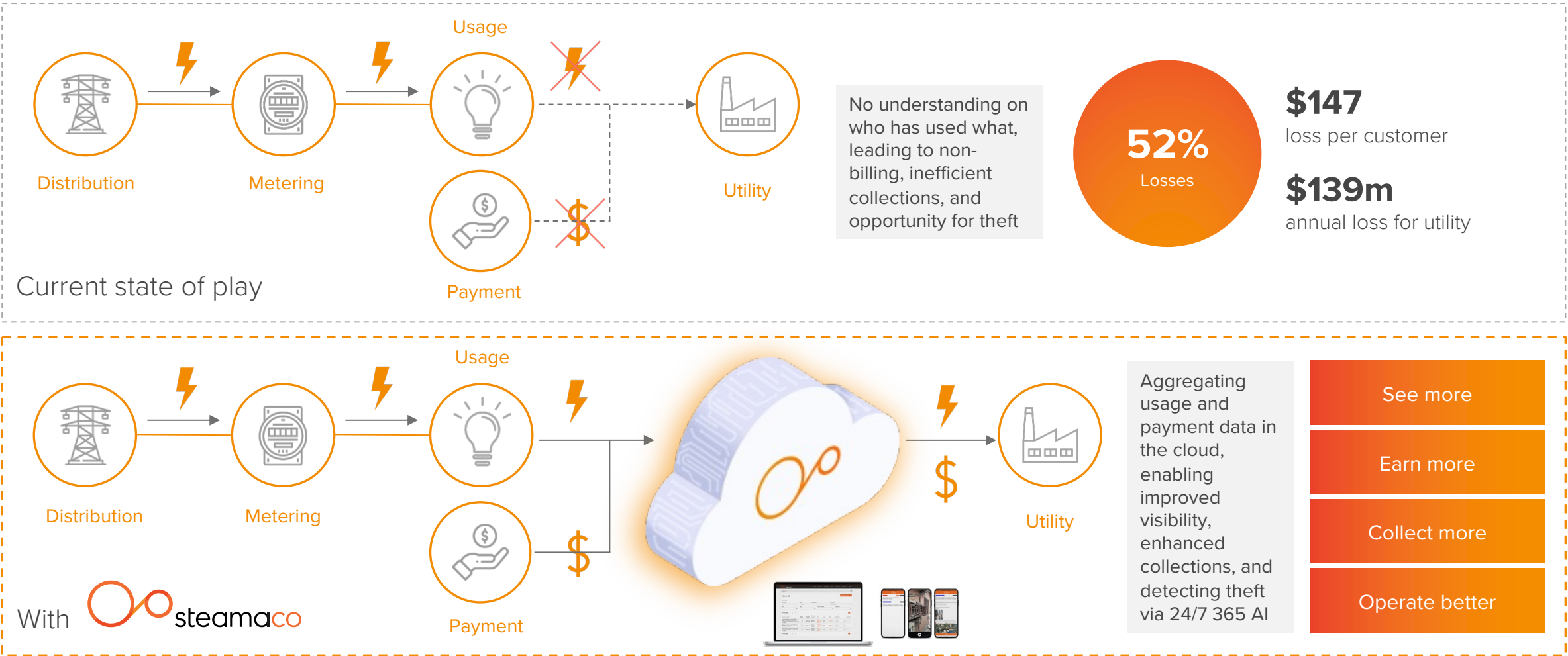
Collecting enough



Efficient enough

Steamaco's solution **collects more revenue.**

Limited infrastructure means traditional solutions don't work.



Helping utilities reduce operational burden with intelligence and automation

Steamaco's classic product provides utilities with insights to reduce Opex and build customer trust

Interoperable solution to reduce Opex, protect assets and build trust with utility customers



Frictionless: API-led solution integrates with existing utility systems



Automation: reduced costs to serve with automated key processes



Alerting: proactive maintenance and customer service



Visibility: complete accountability on usage and payments



Field-force management: improved efficiency of field service trips



Connectivity: low-cost, reliable and self-healing connection services



Protection: power quality monitoring to ensure grid health



Flexible: tariff choice and support for value added OTT⁽¹⁾ services



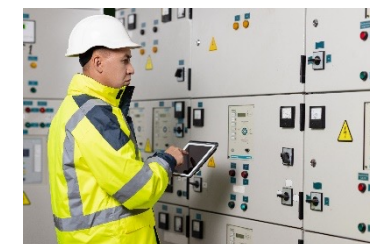
Trusted: better customer service through improved communication

Interfaces and support suitable for each use case

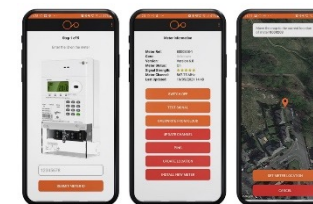
Simple user-interface



Utility HQ



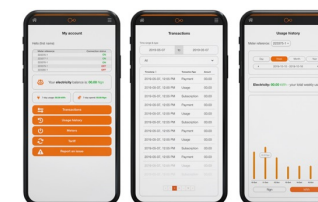
Managed workflows



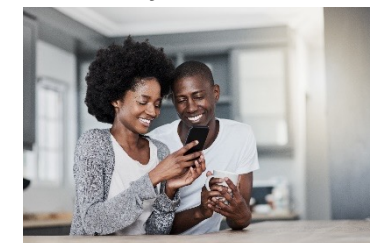
Utility Field Worker



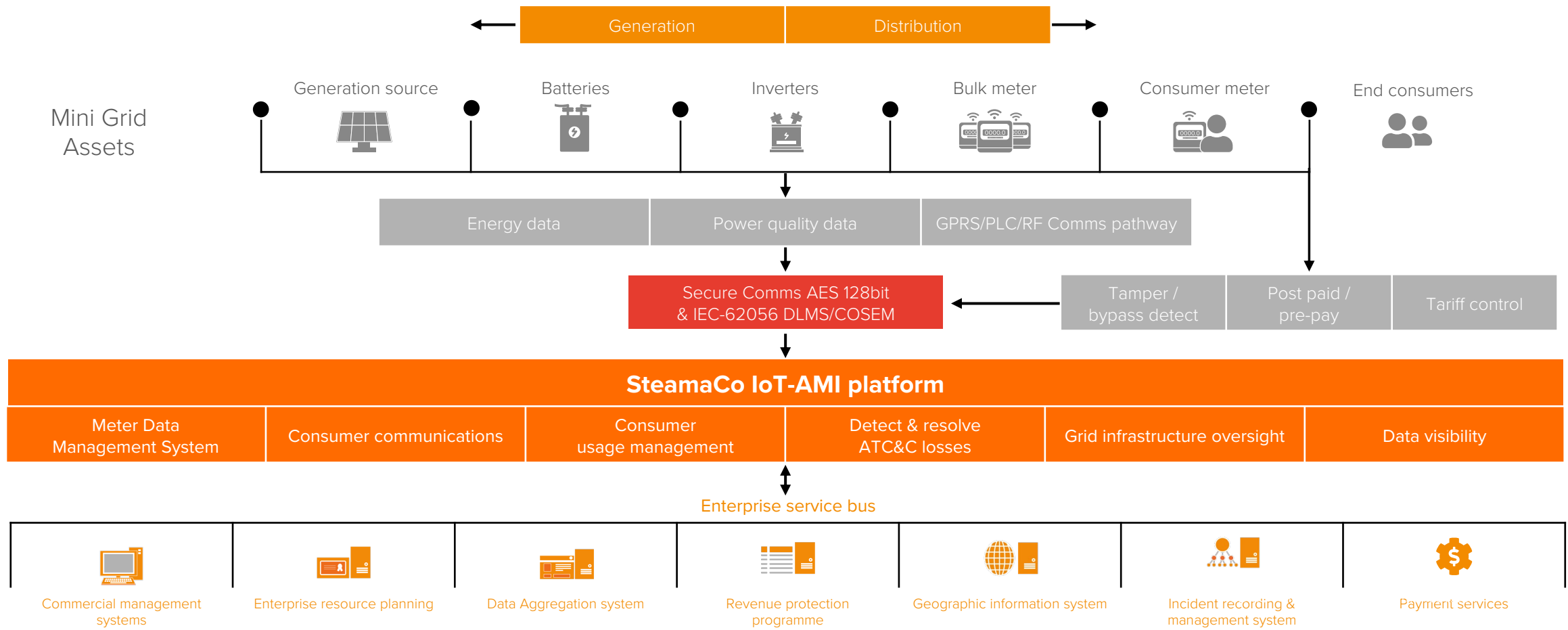
Visibility on usage & costs



Utility customers



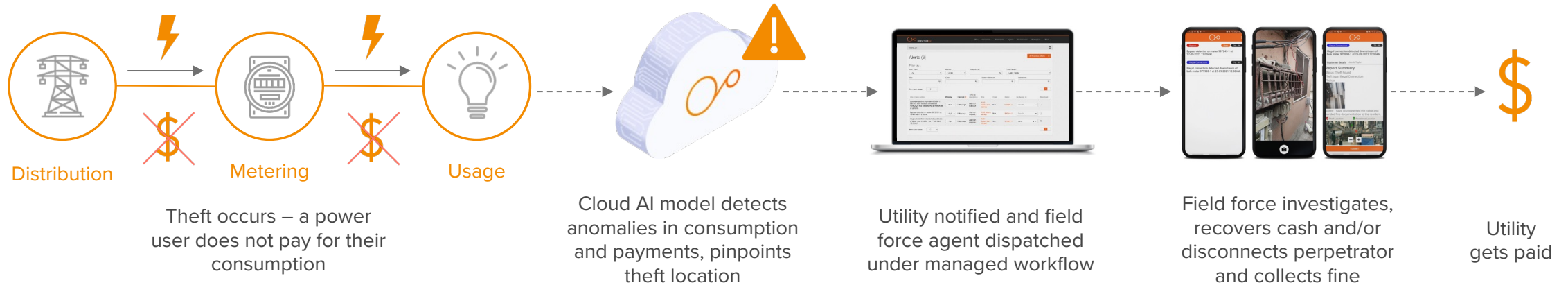
SteamaCo Interoperable IoT-AMI platform for Off Grid



Using AI to detect and resolve theft to increase utilities' revenues

The new product dramatically improves utilities' profitability using simple customer information and hunting for anomalies

AI-driven product enables theft detection, location stamping, and coordination of troubleshooting activities to collect cash



Summary

Flexible. Modular. Interoperable.

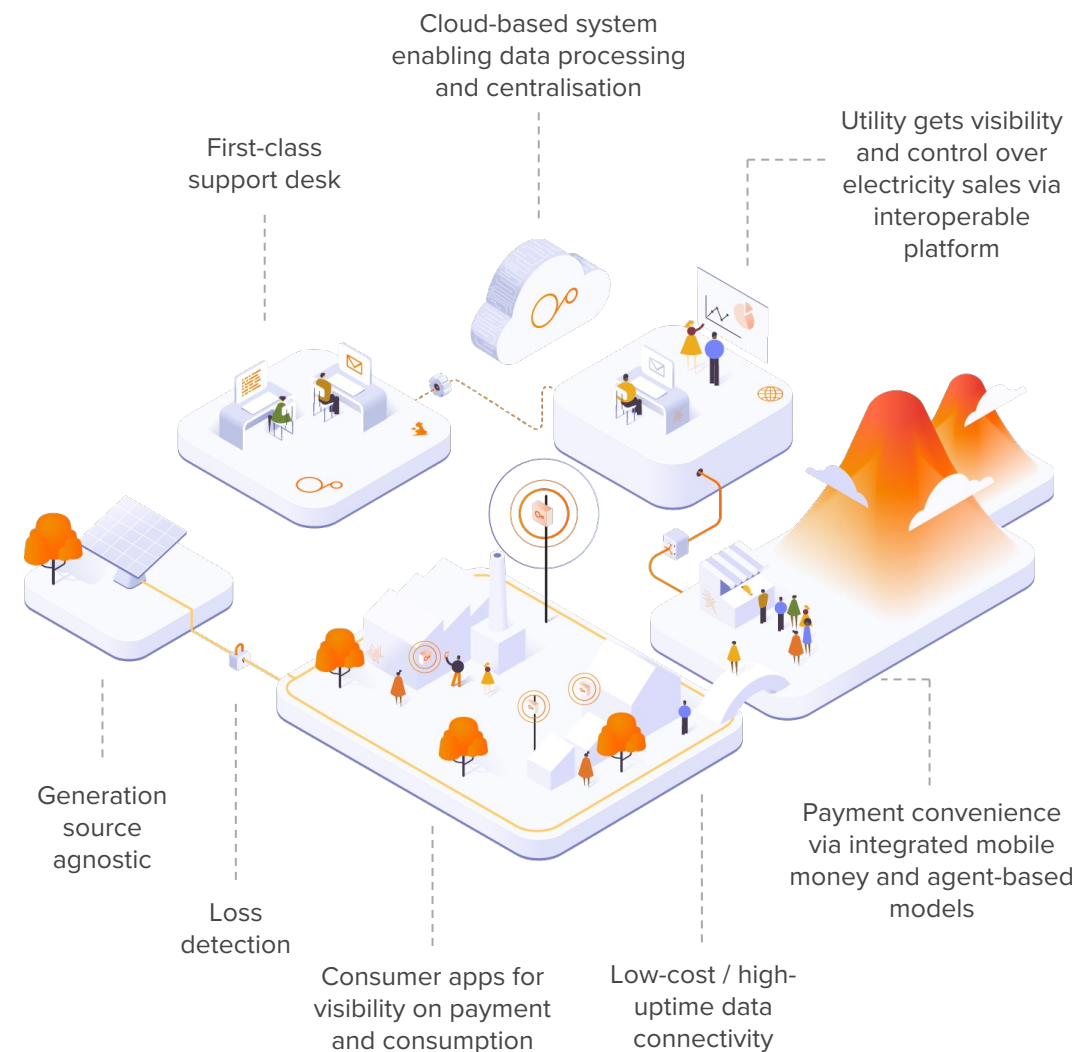
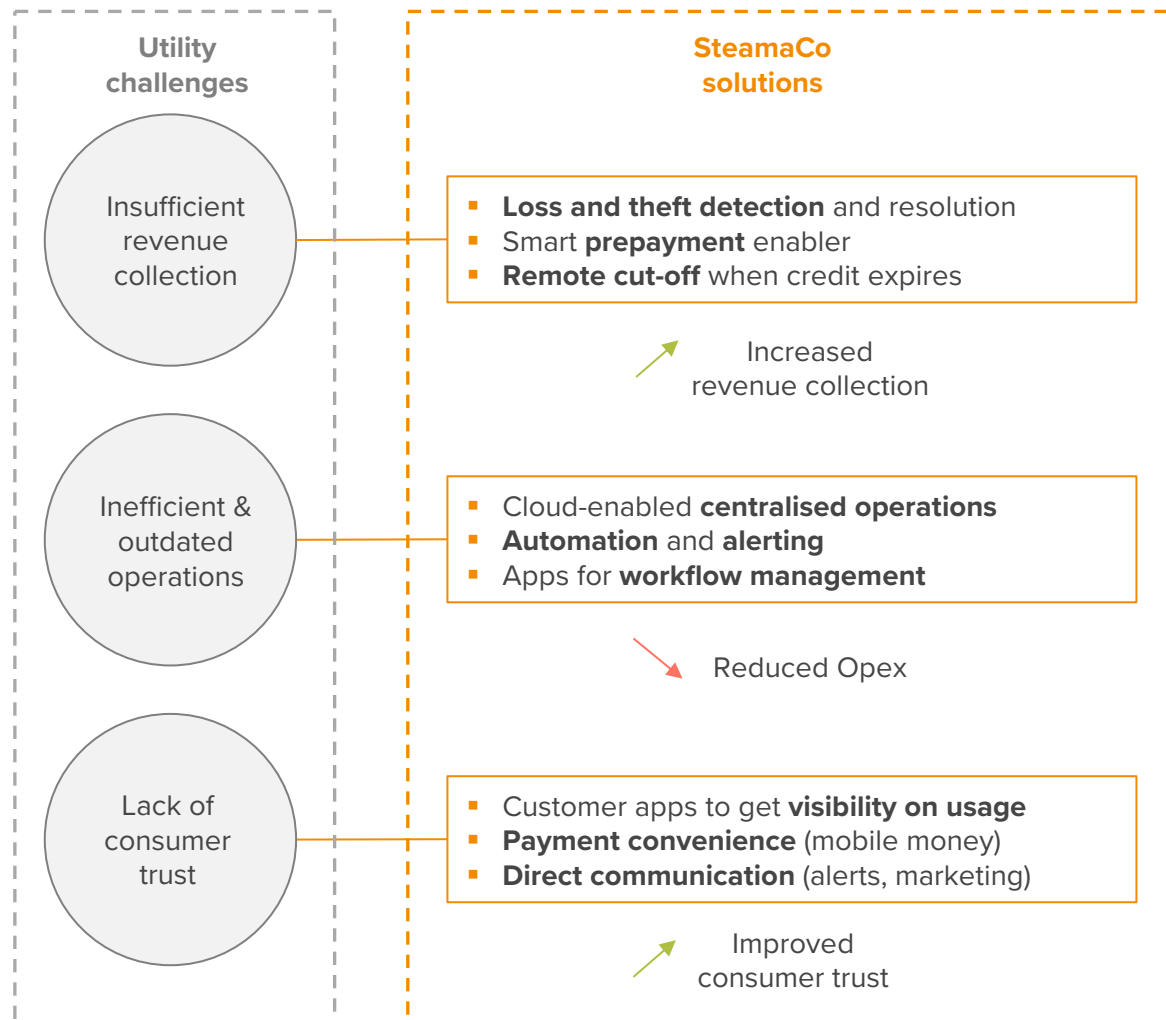


Steamaco IoT smart metering platform



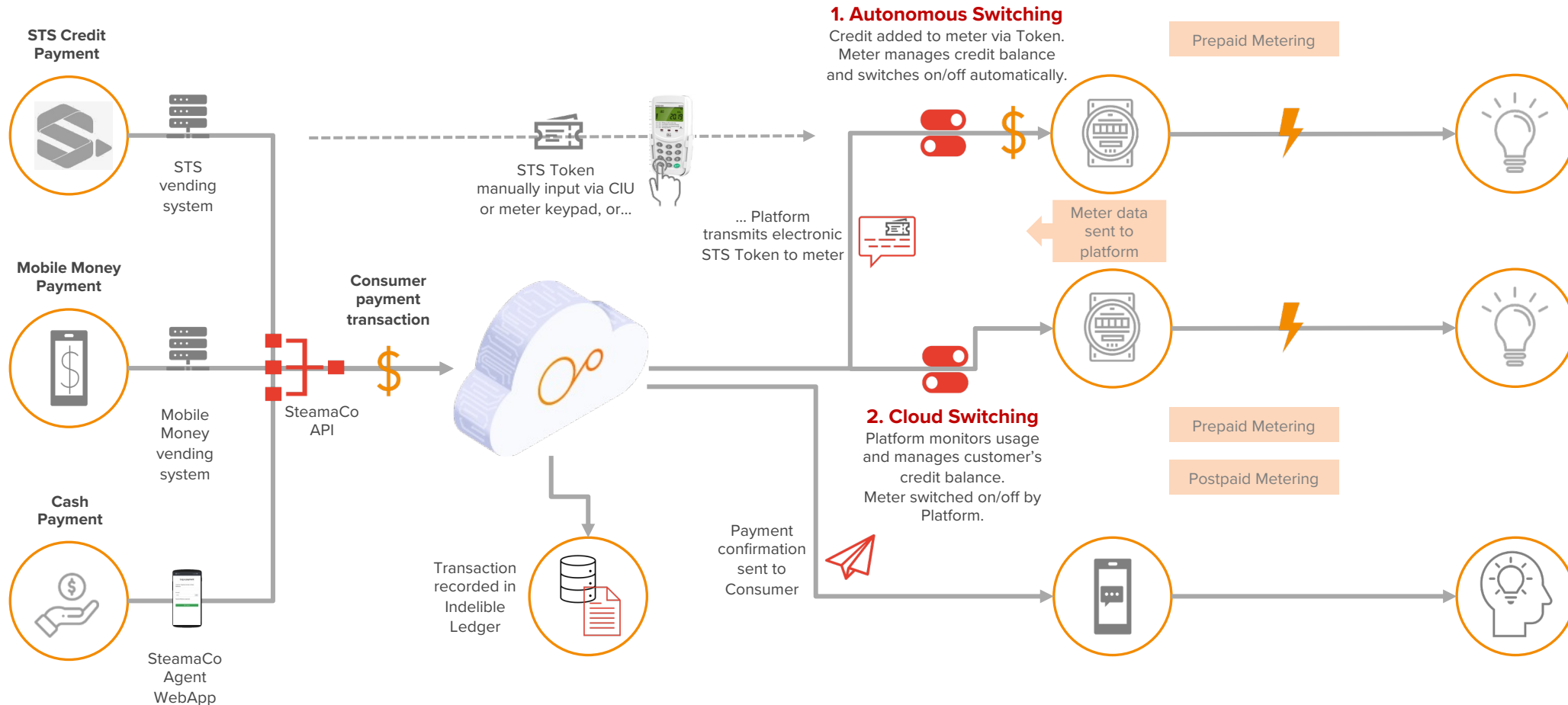
Helping utilities sell electricity profitably with consumer intelligence

The platform overcomes key utility challenges with a unique set of technologies developed for Emerging Markets



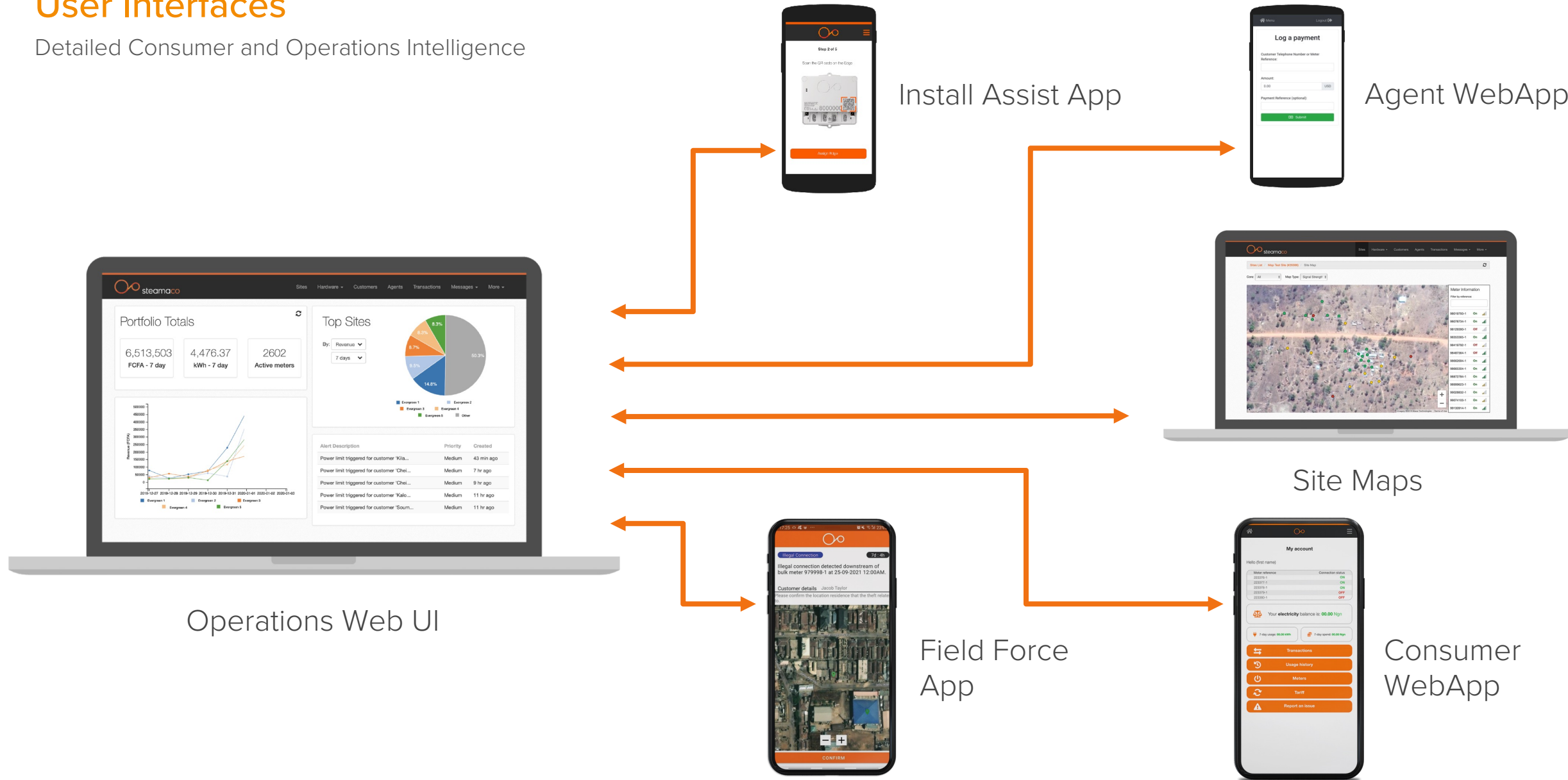
Energy payments and meter switching

- 1 Payment data captured from mobile money, agents, and STS PoS ...
- 2 ... and ingested into the SteamaCo AMI Platform as payment transactions ...
- 3 ... the consumer's credit balance is updated in the platform (also in their meter if it's operating autonomously) ...
- 4 ... and energy supply is switched on and the consumer kept informed.



User Interfaces

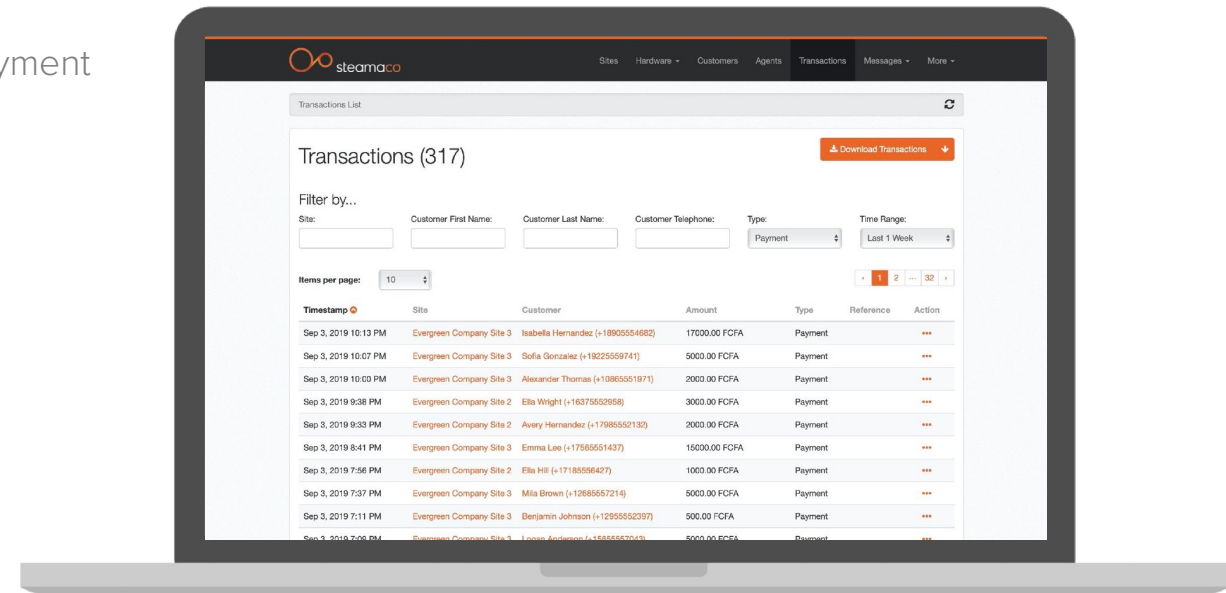
Detailed Consumer and Operations Intelligence



User Interface Capabilities

Easy payment, billing & flexible tariffs

- Prepaid and post paid meter service
- Easy to resolve customer complaints
- Cash and mobile money payments accepted
 - Agent App ensures secure, accurate and convenient payment entries
- Integrates with your chosen payment and financial reporting platforms
- Cloud-based ledger makes it easy to track, audit and reconcile transactions
- Wide variety of pricing and tariff models:
 - Time-of-use pricing
 - Credit bundles
 - Adaptive daily/monthly kWh tiers
 - Subscription plans with or without usage limits
 - Flat kWh pricing
 - Hybrid standing charge and flat kWh pricing



User Interface Capabilities

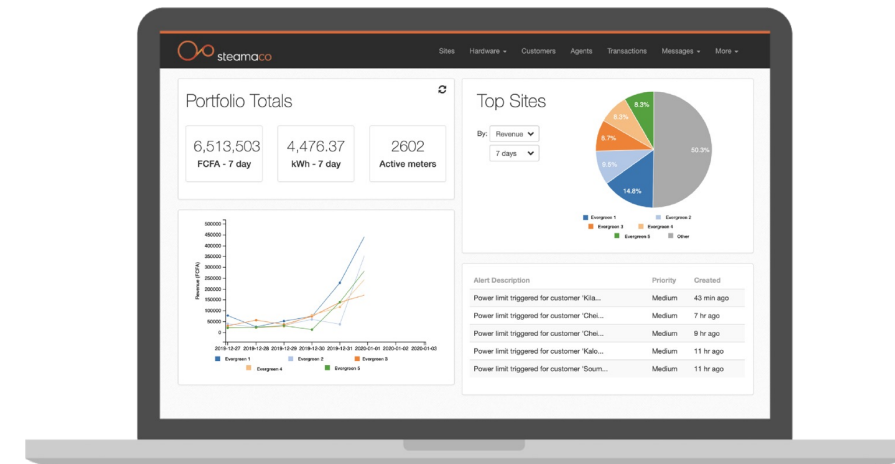
Simplify and automate day-to-day utility actions, maintenance & troubleshooting

Remote operations

- Search, browse, filter, plot and share data from any meter in clear tables and charts.
- Multilingual consumer messages, including in local languages.
- Send auto-alerts by SMS.
- Technicians: get data including meter readings, signal tests, voltage etc.
- Sales agents: process payments and confirm account balances.
- Consumers: receive messages including account balances, low balance warnings etc.

Integrations

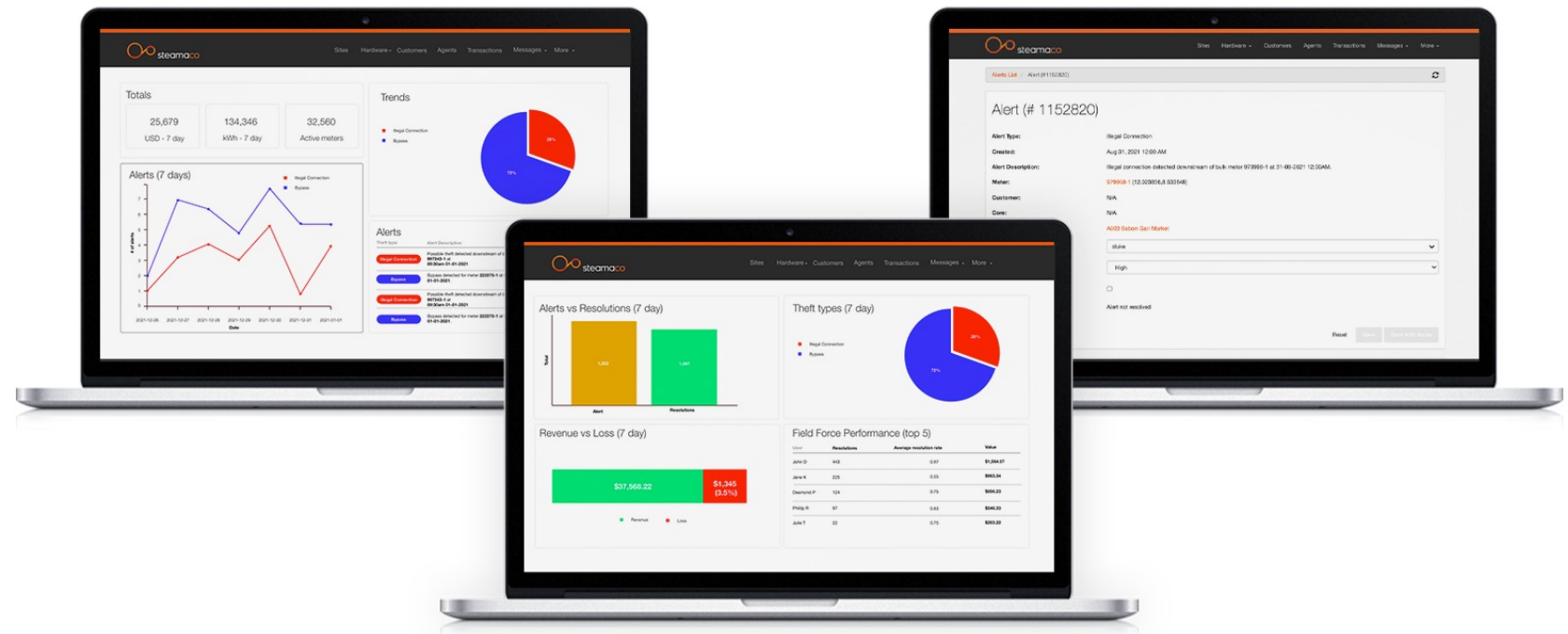
- Customised to work with any metering API on request.
- Can integrate with other compatible (DLMS) smart metering hardware
- Runs on the same mature, secure and successful REST API that operates SteamaCo's smart meters currently in use by utilities in ten African countries.



User Interface Capabilities

Reduce field maintenance costs and technical & consumer losses

- Get automatic alerts for detected losses.
- Detect technical losses and troubleshoot remotely.
- Meter bypass detection.
- Optimise maintenance and service planning to reduce the time spent on site.
- Continual error checking on meter readings.



Security and Data protection

Comprehensive protection built into the platform and the Cloud services



Protection built into our SaaS platform

Permissions-based User Access Control

- Prevents unauthorised access
- Individual users can only view or edit what they are allowed to

Audit logs

- records every user's actions performed on the system

Data archiving

- Granular data older than 90 days is securely archived out of the platform database

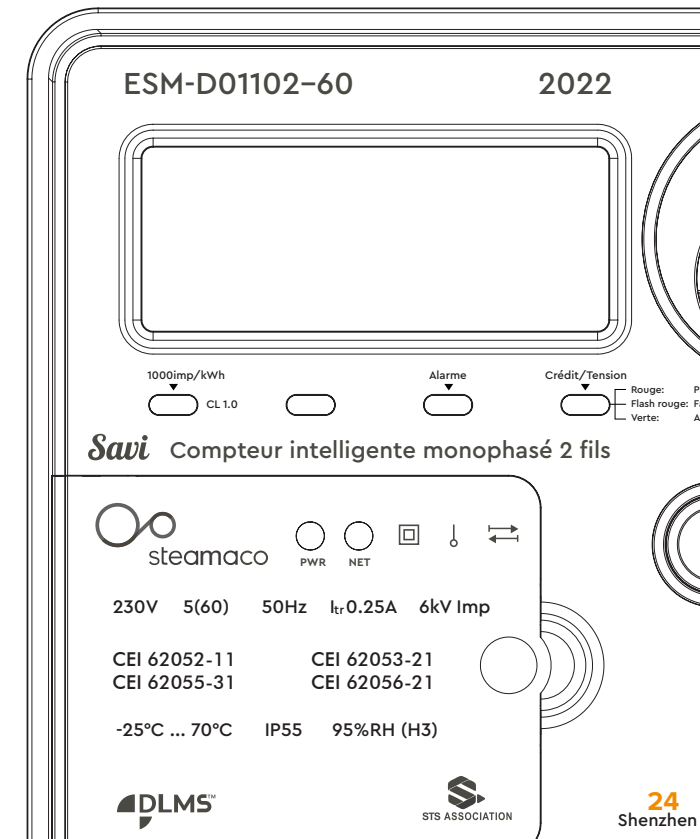
Data privacy

- Each customer can only access their data via their UI logins or their API keys.

Protection provided by Amazon Web Services



Savi Meters – Detailed Overview



Savi smart meters : the product line up

Single Phase : 60A and 100A

Single Phase DIN-Rail : 60A and 100A

Three Phase : 60A and 100A

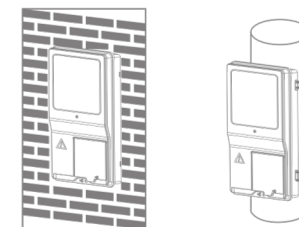
Bulk Meters

- CT-Operated meters for bulk supply measurement: 100A → 3kVA+

Data Concentrator Unit: Available Q4 2022

Key Features and Standards

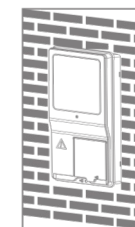
- Metrology : IEC 62052-11, IEC62053-21
- STS : IEC 62055
- DLMS/COSEM : IEC 62056
- Communications : GPRS/4G; Hybrid G3 PLC/RF-Mesh
- Prepaid and postpaid consumer metering
- Configurable current power limits
- Multiple tariffs supported on-board and via cloud
- Option for Customer Interface Unit
- Anti-tamper detect with battery backup
- Fully interoperable with AMI and HES systems via DLMS



Savi smart meters : the product line up

Major Specifications

- Reference voltage (Un) : 230V / 50Hz
- Operational voltage : 70%Un ~ 120%Un
- Operating temperature : -25°C ~ 75°C
- Accuracy class : 1
- Connection types : LNNL (LLNN for DIN-Rail)
- Ingress Protection : IP55
- Hot-swappable communications modules (excl. DIN-Rail)
- Meter enclosures available, wall & pole mount options
 - Single meter box : Savi IP55 meters & Savi DR
 - Small meter box : 2 x Savi DR meters
 - Medium meter box : 4 x Savi DR meters
- Multiple meters per box available factory assembled and pre-wired, including ancillaries (MCBs etc.)



Savi smart meters : Single Phase standard meters

60A, no CIU opt	60A, CIU opt	100A, no CIU opt	100A, CIU opt
	NEMSA		NEMSA – IP54 model

Standard spec	IP55, LCD screen, Extended terminal cover, Comms Module
Optional specs	Keypad, Short terminal cover
Accessories	CIU, Single meter enclosures
Compliance	STS & DLMS certified
Type Approval	NEMSA, UNBS, ANM +
Installation	Wall & pole mount
Production options	Factory assembled packaged options for ‘wire-and-go’
	Incl. ancillaries: MCB, Surge Arrestor, Terminal block, pre-wired



Savi DR smart meters : Single Phase DIN-Rail meters

60A, incl. GPRS comms, CIU res.	60A, incl. PLC/RF comms, CIU res.	100A, incl. GPRS comms, CIU res.	100A, incl. PLC/RF comms, CIU res.
NEMSA	NEMSA: Q422	NEMSA: Q422	NEMSA: Q422

Standard spec	Short terminal cover, LLNN terminals, Built-in communications
Accessory	CIU
Enclosure (IP55)	Small. : 1 – 2 meters (\$ 5.50) Medium : 3 – 4 meters (\$ 20.00) Large : 6 – 8 meters (\$ TBD)
Compliance	STS & DLMS certified
Type Approval	NEMSA, UNBS, ANM, +
Installation	Wall & pole mount (within IP55 enclosure)
Production options	Factory assembled packaged in enclosure for ‘wire-and-go’
	Incl. ancillaries: MCB, Surge Arrestor, Terminal block, pre-wired



Savi smart meters : Three Phase standard meters

60A, no CIU opt	60A, CIU opt	100A, no CIU opt	100A, CIU opt
	NEMSA: Q422		NEMSA

Standard spec	IP55, LCD screen, Extended terminal cover, Comms Module
Optional specs	Keypad, Short terminal cover
Accessories	Single meter enclosure
Compliance	STS & DLMS certified
Type Approval	NEMSA, UNBS, ANM +
Installation	Wall & pole mount
Production options	Factory assembled packaged options for ‘wire-and-go’
	Incl. ancillaries: MCB, Surge Arrestor, Terminal block, pre-wired



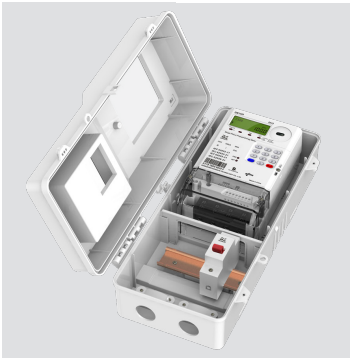
Savi Bulk smart meters : CT Operated meters to measure bulk supply

6A, Class 0.5s	10A, Class 0.5s	IP55 Enclosure	CT clamps
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NEMSA

NEMSA

Standard spec	IP54, Extended terminal cover, modular comms bay
CT Clamps	Quoted to suite current rating • 100 – 600A @5A • 600A+ @ 10A
Application	Single phase line current measurement & Three phase line current measurement, for: • Distribution Transformers • Line Totalisers • Bulk current supply measurements
Accessories	Single meter IP55 enclosure
Compliance	DLMS certified
Type Approval	NEMSA, UNBS, ANM, +
Installation	Wall & pole mount



Savi DR2 & DR4 smart meter packages : Deploy meters faster

Savi DR2 - GPRS	Savi DR4 - GPRS	Savi DR2 - Hybrid	Savi DR4 - Hybrid
2 x Savi DR	4 x Savi DR	2 x Savi DR	4 x Savi DR

Save time, money and install meters faster with multiple DIN-Rail ‘meter boxes’, pre-wired & factory assembled

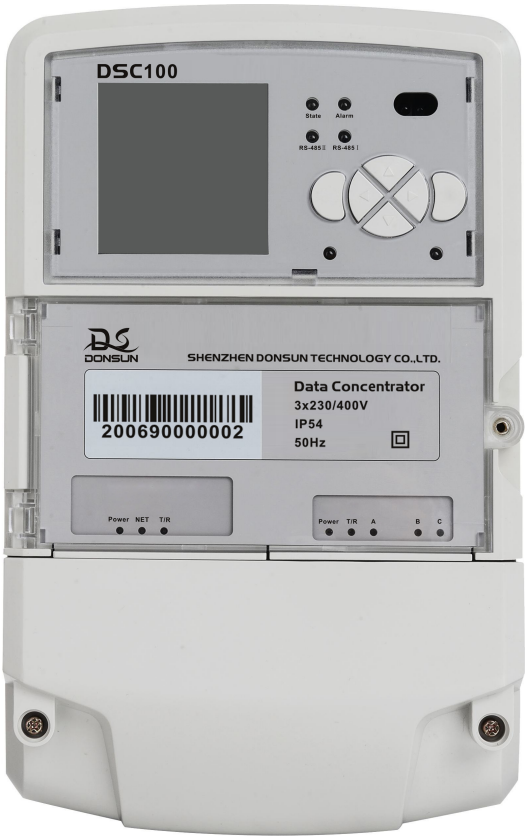
- **Applications for the DRx packages :** Suitable for grid, sub-grid and off-grid metering
 - Option to add a CIU per meter
- **Communication options:**
 - Individual Comms: each meter in the box is fitted with GPRS or Hybrid communications
 - Comms Router: a primary meter routes comms between itself & all secondary meters and the AMI – available in 2023
- **Factory-fitted ancillaries:** MCB per box, MCB per meter, surge arrester
- **Installation:** wall or pole mounted, IP55 enclosure for 2 or 4 connections



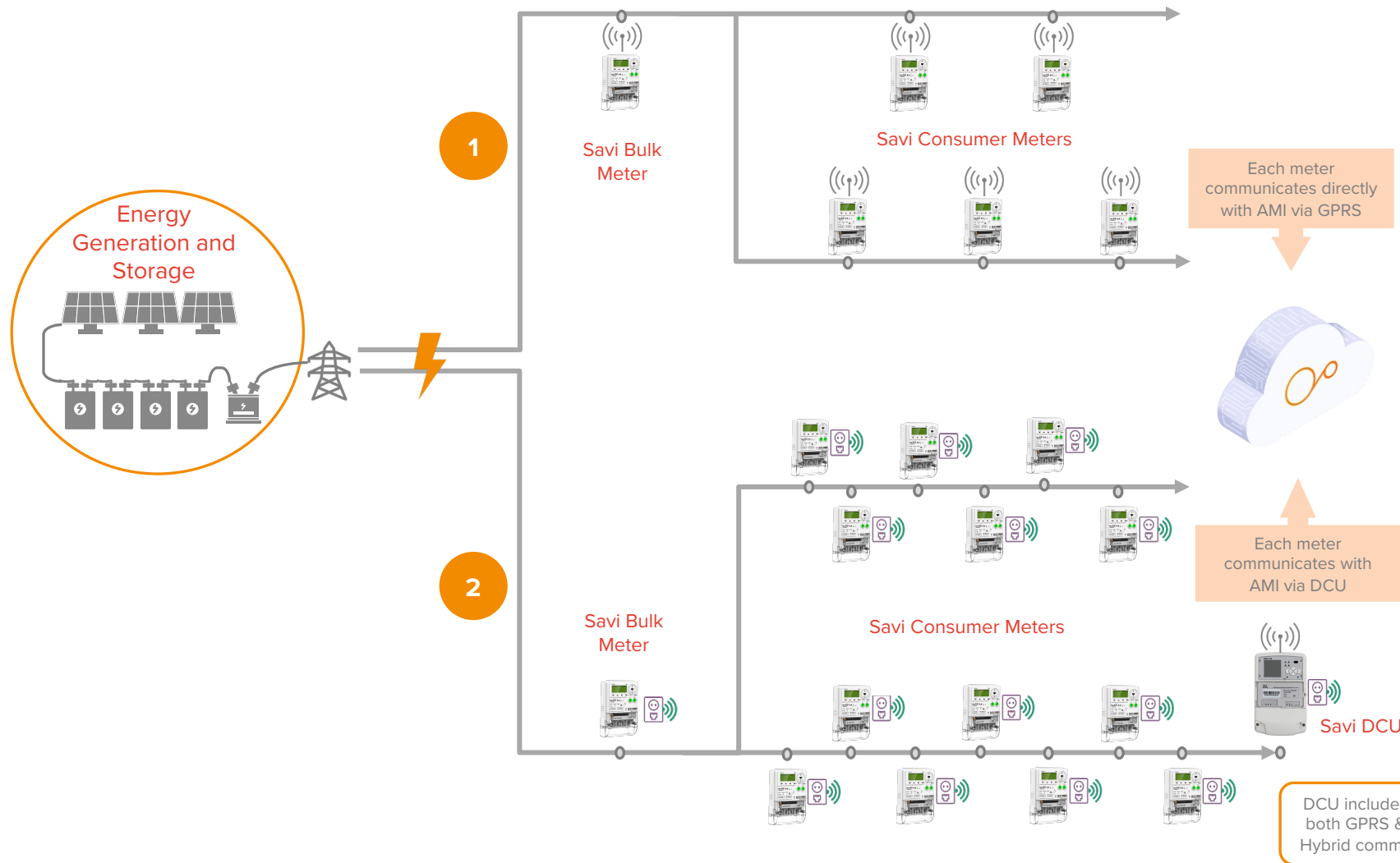
Savi smart meters : Data Concentrator Unit

Savi DCU	8dBi GSM Antenna	27dBi GSM Antenna	IP55 Enclosure
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Function	Communication gateway and intelligent at-grid data management
AMI communication (WAN)	2G-GPRS / 4G or TCP/IP
Meter communication (LAN)	Hybrid G3 PLC/RF
GSM range extension	High-gain omni & directional external GSM antennas
Communication standard	DLMS/COSEM (HDLC and TCP/IP profiles)
Specifications	• 500 meters • 1km G3-PLC
Memory	• Meter data storage:
Installation	Wall & pole mount within IP55 enclosure
Availability	Q4 2022



Savi meter communications : GPRS and Hybrid applications



1

Meters with GPRS communication boards

Requirements:

- One SIM card per meter
- Data plan per SIM card

Benefits:

- Meters communicate with AMI independently of each other
- Ideal for single-meter applications or nano-grid sites with few meters (< 20)

Operating Considerations:

- Large mini grids: Cost and effort for managing a very large number of SIM cards and data plans for each SIM

2

Meters with Hybrid G3-PLC/RF communication boards

Requirements:

- Data Concentrator Unit(s) on site
- One SIM card per DCU
- Data plan per SIM card

Benefits:

- Mesh communications between meters and DCU via PLC or RF
- Ideal for mini grid sites with many meters (100's, or 1,000's)
- Lower data cost per meter per month

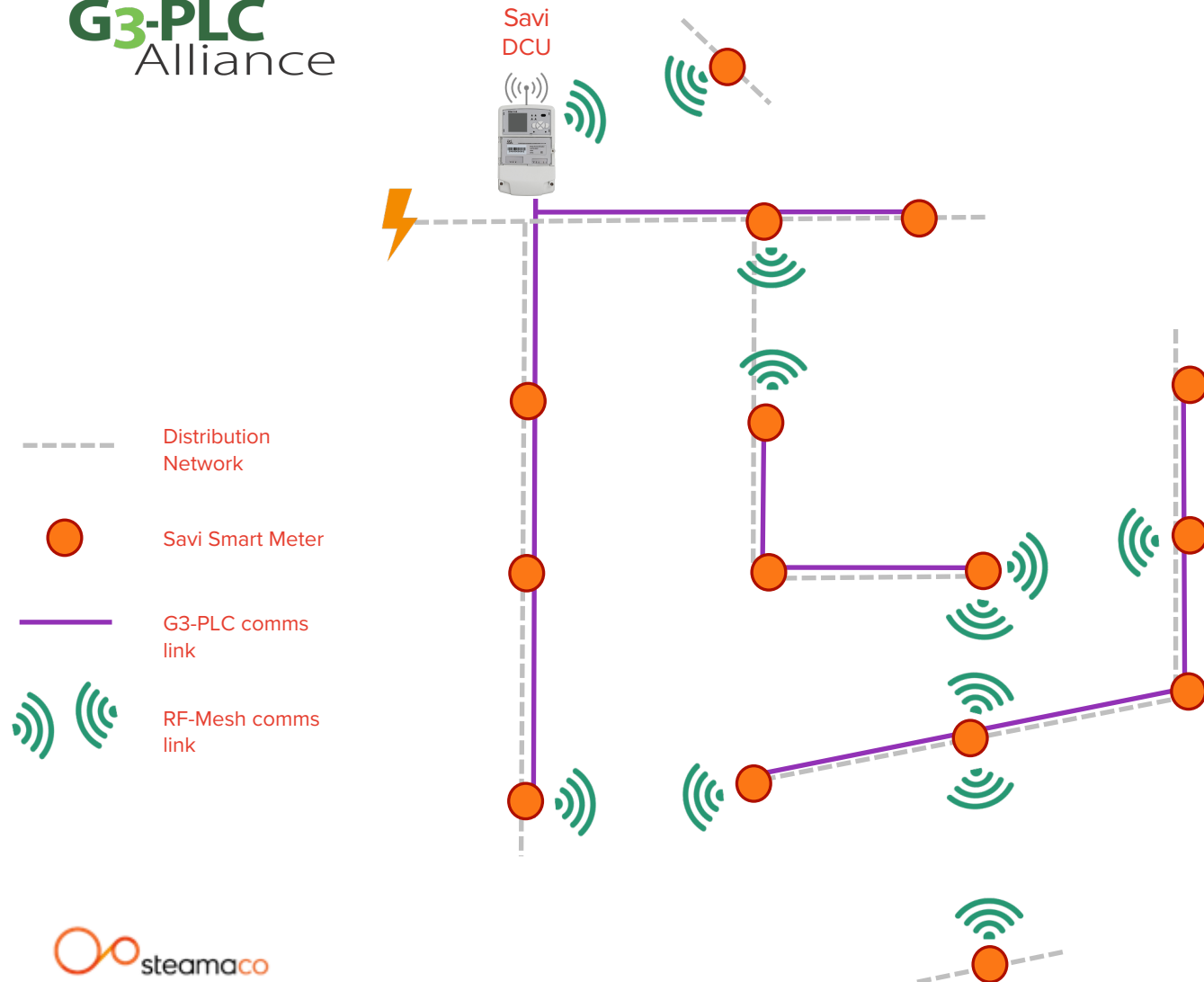
Operating Considerations:

- Cost and effort for managing small number of SIM cards and data plans (1 per DCU)
- Outlying meters situated on site fringes can use GPRS if too far out of PLC/RF mesh comms range with DCUs

Savi meter communications for mini-grids – Hybrid G3-PLC/RF

Hybrid G3-PLC/RF – robust, mesh communications designed for reliability, even in challenging environments

G3-PLC
Alliance



- **Secure:** data communications protected by AES-128bit encryption
- **Interoperable:** supports **DLMS/COSEM** & IEEE 802.15.4 & Ipv6
- **Reliable:** G3-PLC Alliance field trials proved **98% - 100%** daily read rates (uptime)
- **Robust:** Uses both **G3-PLC** and **RF-Mesh** (868MHz) to maximise coverage & connectivity across any distribution network
- Created by the G3-PLC Alliance for utilities world-wide
- Ensures successful communications in all environments and types of distribution grids: both rural and urban
- Automated and dynamic operation:
 - Messages transfer between DCU and target meter via a mesh network of comms links
 - The best media (PLC or RF) is selected for each individual comms link in the mesh network
 - Automated channel selection during network setup
 - Dynamic adjustment based on PLC or RF link status (e.g. if at one time, PLC fails for a link, then link switches to RF)

Confidential

Savi DR meters: Factory assembled, multi split-meter solutions for distribution grids

Ensuring fast, safe and secure installations of single-phase smart meters, with CIU option

1

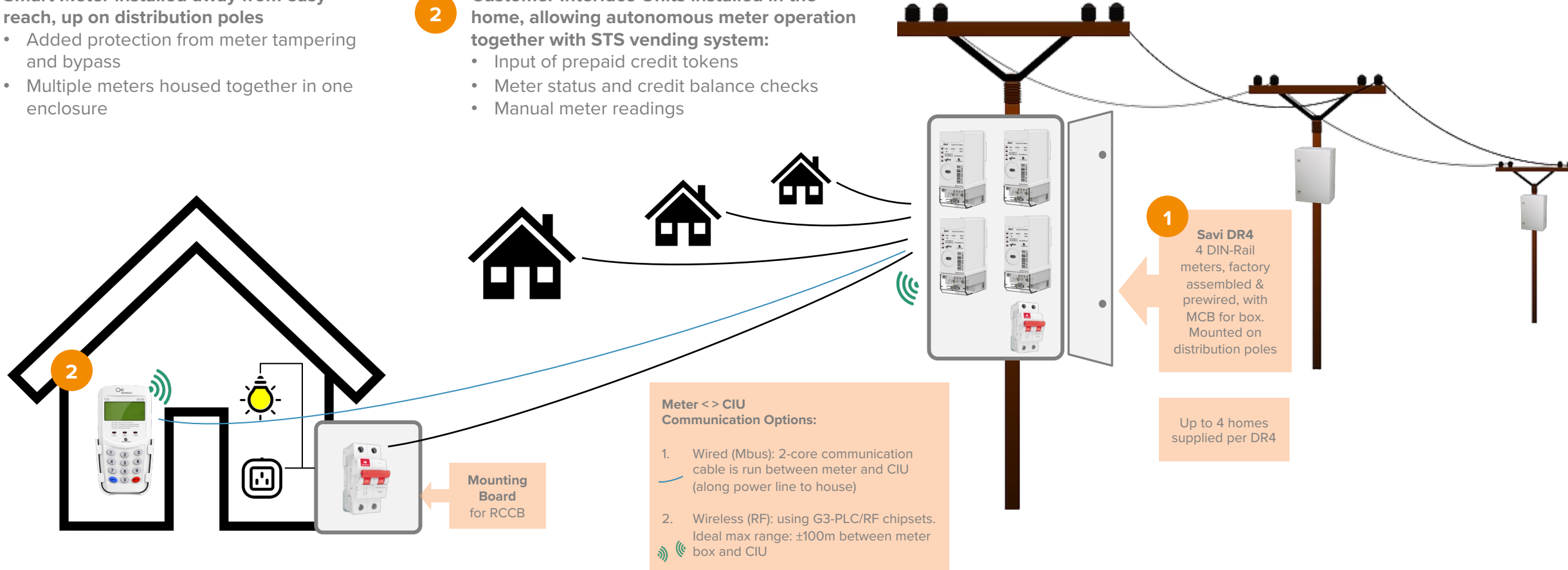
Smart Meter installed away from easy reach, up on distribution poles

- Added protection from meter tampering and bypass
- Multiple meters housed together in one enclosure

2

Customer Interface Units installed in the home, allowing autonomous meter operation together with STS vending system:

- Input of prepaid credit tokens
- Meter status and credit balance checks
- Manual meter readings



Steamaco Support Services



SteamaCo support



- 24/7 online help
- 7 day per week UK-based support team
- Remote support for onboarding & higher level troubleshooting
- Technology that diagnoses, alerts & fixes comms issues
- Remote site monitoring to identify comms/power quality issues



Collaborating for success

- Collaborate on technical and commercial specifications
- SteamaCo supports with in-country compliance with regulators
- Communications and site planning to optimise product performance
- Onboarding and training to support roll out
- Go-live support
- IoT feature development
- Experienced engineers available to assist in the implementation with software integrations

